FINAL PROJECT

Trainer Progression App

Next week’s workouts drafted from last week’s logs. The trainer approves every one.

Yadan Taino and Justin Day

Who we are

CLIENT SIDE AND DESIGN

Yadan Taino

Built what clients and trainers touch: chat and voice logging, the Workout Logger agent, the progression rules, the Trainer Assistant check-ins, and the app’s design.

BACKEND AND DEPLOYMENT

Justin Day

Built the engine room: the API and database, the Progression Planner agent, trainer sign-in, and the live deployment on Render and Vercel.

The job: personal trainer

Why we chose it: a trainer we know rebuilds every client's program by hand each week. Of the four core duties, two repeat for every client, every week.

Assess abilities and goals

Intake, history, and physical assessment

Create and adjust programs

Individual plans, updated as clients progress

Our focus

Monitor client progress

Compare logged results and spot trends

Our focus

Teach technique and safety

Live observation and correction

Sources: O*NET 39-9031.00; U.S. Bureau of Labor Statistics, Fitness Trainers and Instructors

The pain points: slow, repetitive, easy to miss

SLOW

Rebuilding every program, every week

She copies each client's program in Google Sheets and edits it lift by lift.

REPETITIVE

The same rule, applied by hand

Reps go up to 12, then weight once it feels easy. The same math for every lift and client.

EASY TO MISS

Signals get lost

Pain mentioned in a text slips by, and a client who quietly stops logging goes unnoticed.

It's not just her. Trainers using Trainerize report the same manual editing. Its AI builds single workouts, not multi-week programs.

Source: our trainer interview and Trainerize market research (r/personaltraining, Capterra, Software Advice)

What we made easier, and how

Today

- Clients open a spreadsheet to see workouts
- Trainer copies last week’s program
- Trainer adjusts every weight by hand
- Pain mentioned in passing is easy to miss
- A client drifting away goes unnoticed

With our app

- Clients log by chat or voice, with no account
- Next week is drafted from reps and effort
- Trainer reviews each change and its reason
- Pain is flagged and that lift is held
- Trainer sees who needs a check-in, message drafted

Target: plan a client’s week at least 50% faster than in Sheets. Not yet timed with our trainer.

Architecture: AI reads, code decides the numbers

Client on phone
Next.js on Vercel, private link, no account

Trainer dashboard
Next.js on Vercel, Supabase login, approvals

FastAPI backend on Render: every action checked for permission

Workout Logger
AI, Sonnet 5.5: turns chat into sets

Progression Engine
Code: the only source of weights and reps

Progression Planner
AI, Haiku 5.5: flags safety conflicts

Trainer Assistant
AI, Sonnet 5.5: drafts check-ins

Supabase: PostgreSQL and trainer login

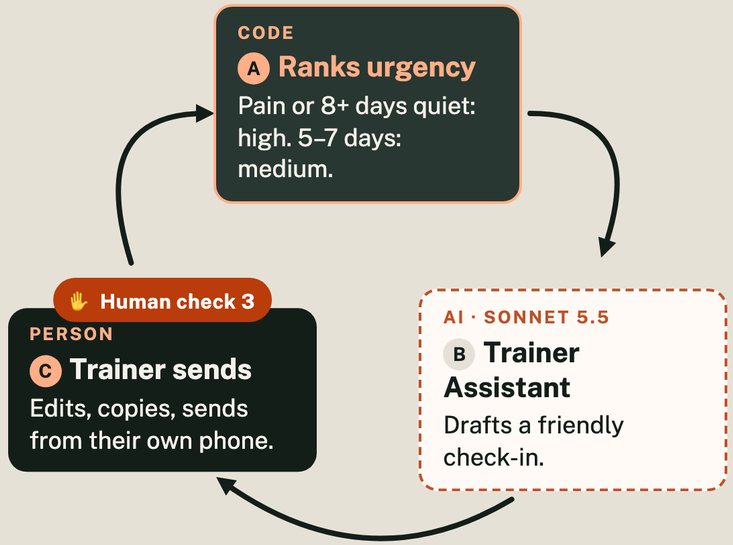
Agent workflow: one weekly loop, three human checkpoints

Every workout goes around the loop once. The AI reads and drafts. Code sets the numbers. People decide.



CHECK-IN LOOP

Runs whenever the trainer opens the dashboard



Guardrails

- AI output is checked: known exercises, reps ≤ 200, weight ≤ 2,000 lbs. One retry.
- Client words are passed in as data, so they can't give the model orders.
- If a model fails, the client's text stays saved, a saved log is never lost, and check-ins fall back to a plain draft.

Person · AI agent (reads and drafts only) · Code (sets the numbers) · Human checkpoint

How one workout moves through

PERSON

1. Client logs

“Bench 135, last set only 8, shoulder pinched”

AI

2. Workout Logger

Pulls out sets and a pain flag. Saves nothing.

PERSON

3. Client checks and sends

Fixes any number, rates effort, taps Send to Trainer. Only this is saved.

CODE

4. Progression Engine

Easy: reps rise to 12, then weight. Hard, missed, or pain: held.

AI

5. Progression Planner

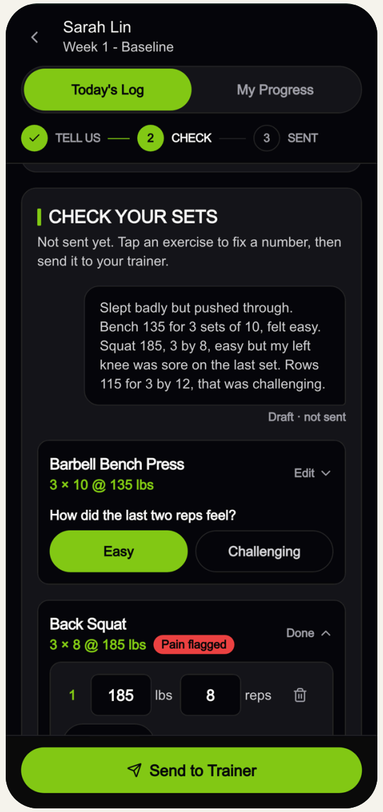
Checks injury notes. Can flag, can’t change a weight.

PERSON

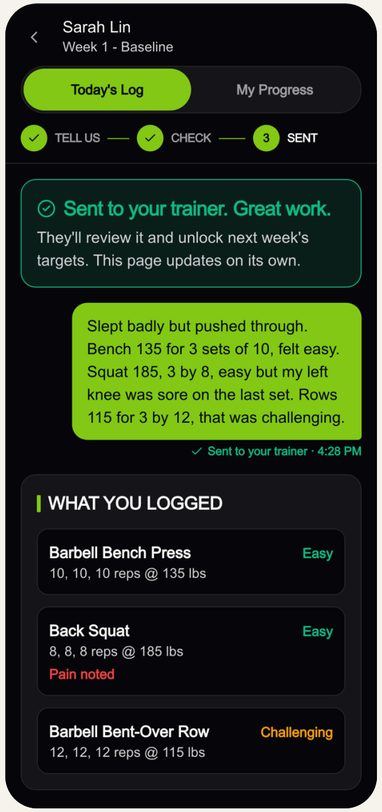
6. Trainer approves

Reads the client’s own words, edits or approves. Only then does the client see it.

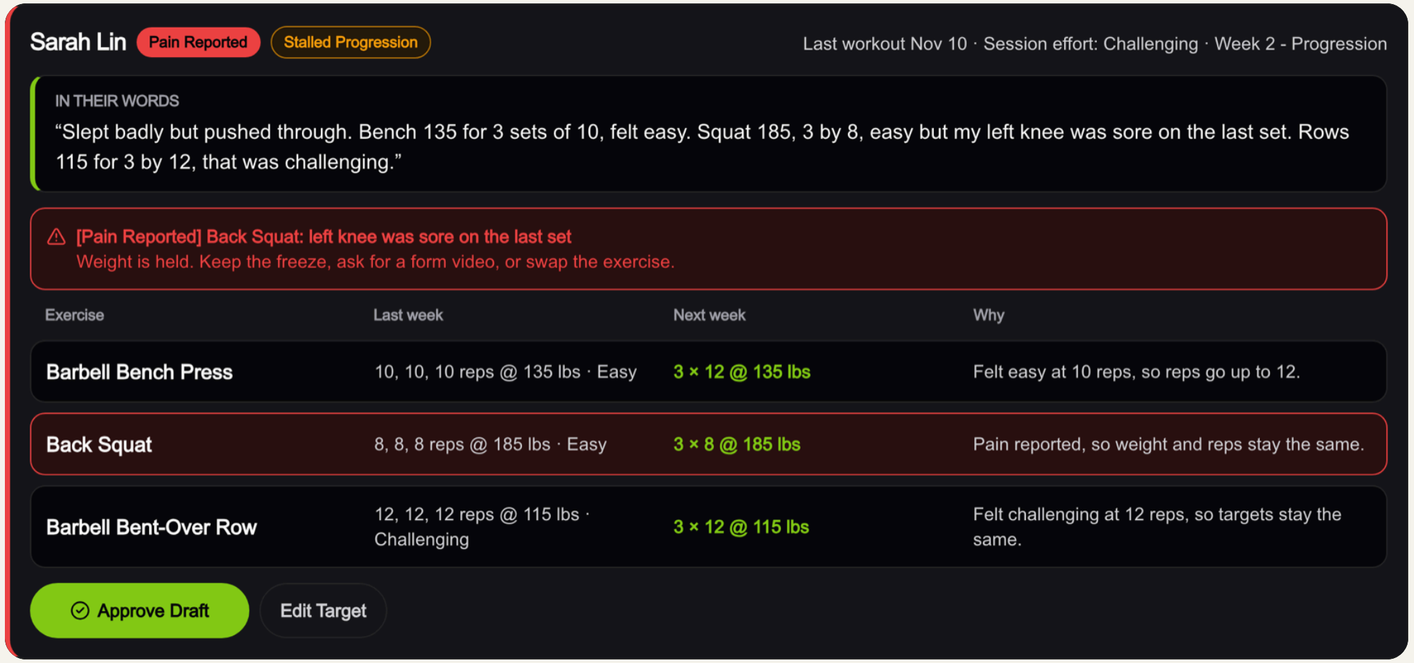
What we built: the whole loop, live

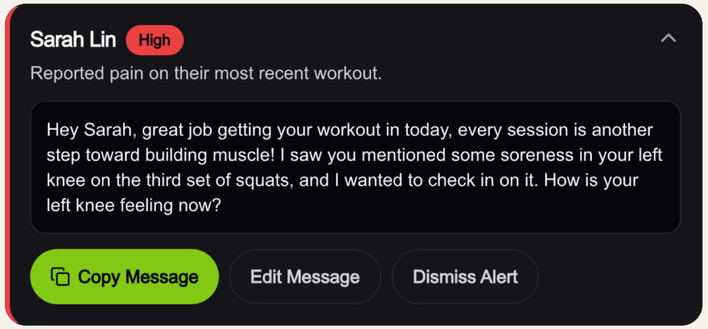


1. Client checks sets



2. Sent to the trainer





3. Trainer approves

Reads her words. Knee pain held the squat. One tap to go live.

4. Check-ins drafted

For clients who need attention.

Screens from the live build, with sample client data

Where it stands: live and tested end to end

trainer-agent-app.vercel.app

What we built	Status
1. Client logging by chat or voice	Done
2. Progression engine and trainer approval	Done
3. Trainer Assistant check-ins and roster	Done
4. Live on Vercel and Render	Live

Final live test on production: 9 of 10 checks passed.

Next

- Log every AI call for review
- Let trainers revoke a client link
- Test on a real iPhone and Android
- Time a real planning week with our trainer